

MICCA Pilot Projects: Identifying climate-smart agriculture practices with smallholder farmers in East Africa

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- Background and objectives of the MICCA pilot projects
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 - Identifying the sets of climate-smart agriculture (CSA) practices
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MICCA Pilot Projects

The **MICCA pilot projects** aimed to:

- **Integrate CSA practices into on-going development activities** to test and demonstrate the synergies and trade offs between food productivity, resilience and GHG emission reduction
- **Provide quantifiable evidence** on CSA to farmers, national and local decision makers and international organizations and donors



MICCA Pilot Projects

Putting climate-smart agriculture into practice

Programme: FAO MICCA Programme

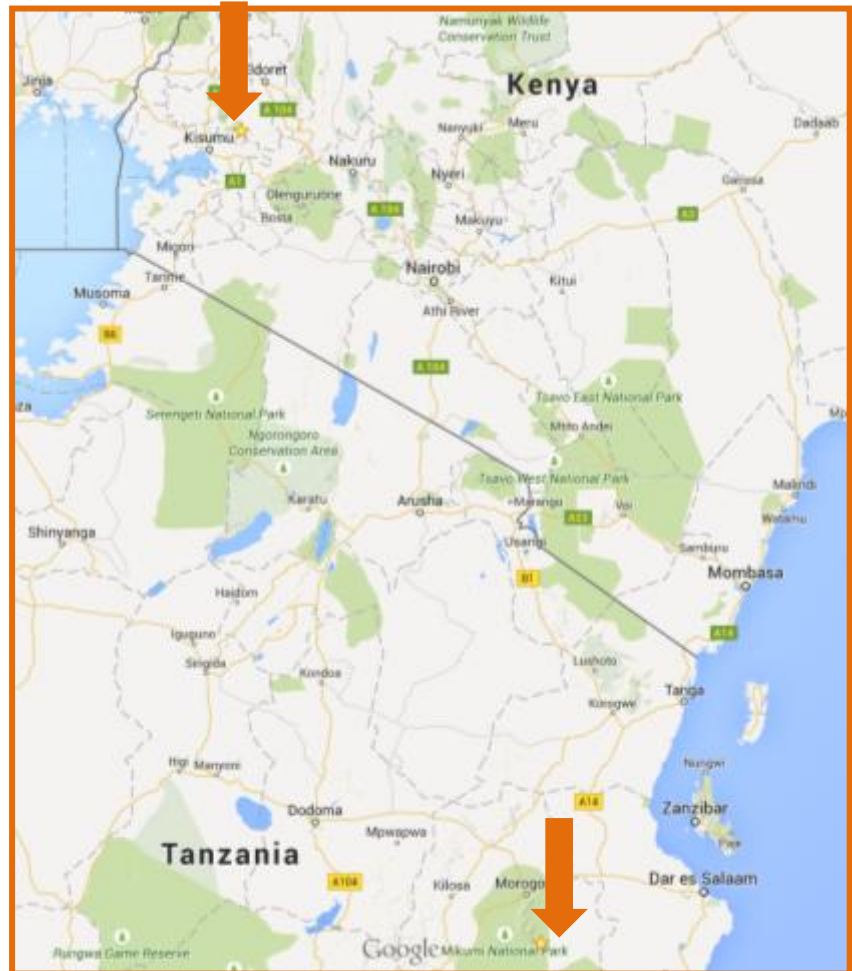
Partners: ICRAF, EADD, and CARE

Timeframe: Jan 2011 to Dec 2014

Donor: The Government of Finland

Locations:

- Kaptumo, Western Kenya, Kenya
- Uluguru Mountains, Morogoro district, Tanzania

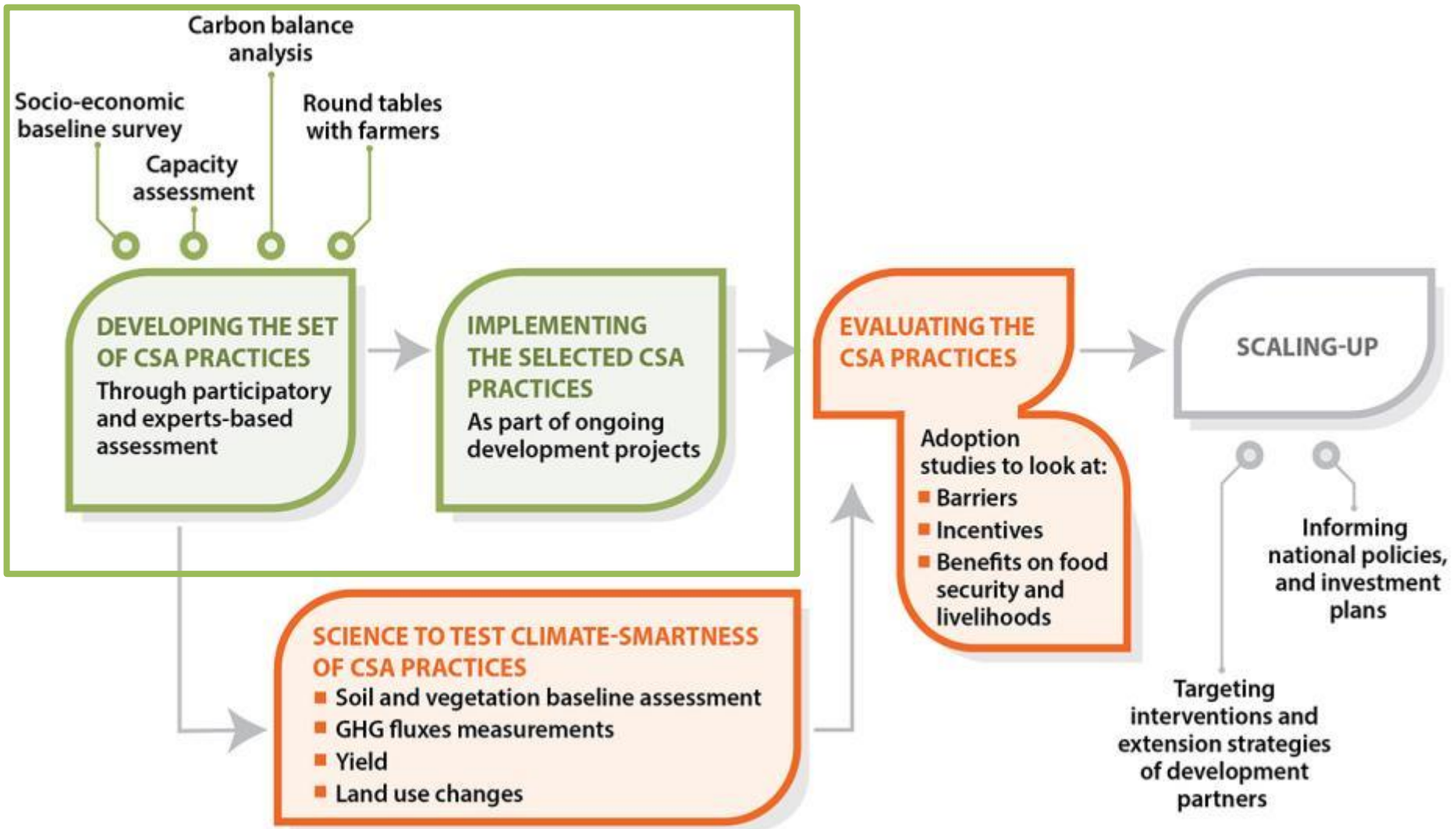


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www.fao.org/climatechange/micca

Approach



Source: Rioux *et al.*, FAO, 2015 (in prep.)



Situation Analysis

- Socio-economic baseline:
 - Representative households survey
 - To identify farming practices, climate risks, socio-economic conditions (inc. gender role, availability of/access to labour and land)
- Capacity needs assessment:
 - Multi-levels capacity assessments at national, district and project
 - To identify farmer needs and policy and institutional environment
- Carbon-balance analysis:
 - To identify the mitigation potential of current and future farming scenarios
- Consultations with farmers:
 - To discuss on different suitable practices and gather their perceptions and preferences



MICCA Pilot Project in Kenya

Integrated crop-livestock system of Western Kenya

Improving milk yield and income of dairy producers while reducing the climate change “footprint” of dairy production system



Kenya: Baseline Results

- **Livestock feeding practices are evolving:**
 - natural pasture all year, only grazing 34%
 - mainly grazing with some stall feeding (9 months pasture) 50%
 - mainly stall feeding and some grazing 15%
 - zero grazing (stall feeding only) < 1%
- **Women** are involved in 50% of decisions on agricultural practices and management, but less in regard to livestock.
- **Farmer interests** in participating in the project was to gain access to loans 37% and improve income 24% (only 4% to gain knowledge).



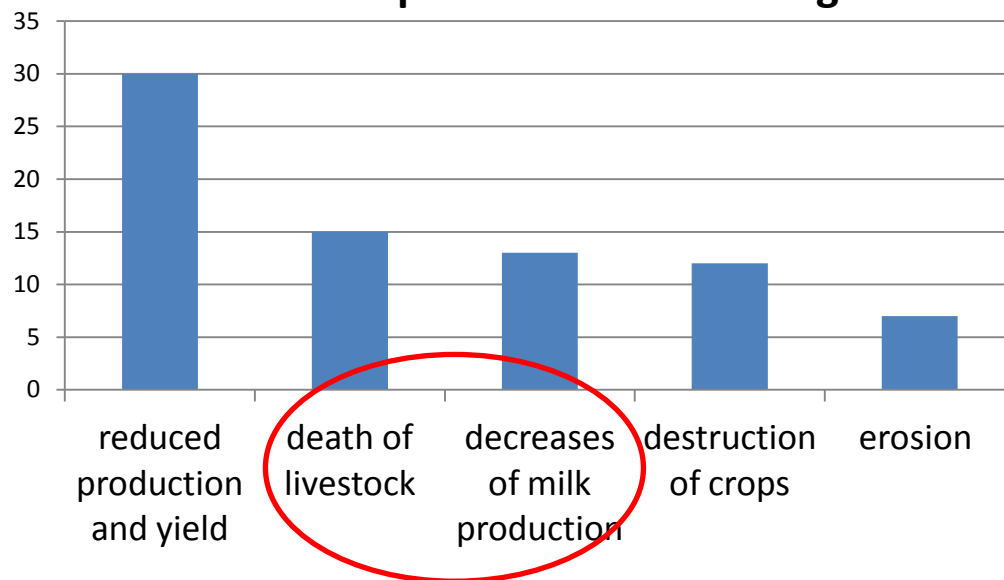
Kenya: Baseline Results

Main changes in climate:

- more erratic rainfalls
- a longer dry season
- rivers drying up
- problems with watering cattle
- decreases in soil fertility

→ Only 11% indicated no change

Impacts of climate change



Coping Strategies

Coping Strategies	%
Building terraces to avoid erosion	10
Reducing herd numbers and improving milk production of smaller herds	10
Changing the type of crops cultivated	8
Changing planting practices	7
Building protective sheds for livestock	7
Growing feed	5



Kenya: Set of CSA practices



Practices	Sub practices or species
Improved fodder production and feed conservation	Napier grass
	Rhodes grass
	Fodder sorghum
	Lucerne (Alfalfa)
	Cow kandy
	Columbus grass
	Dolichos lab lab
	Hay and silage making
Agroforestry/ tree planting/ tree nurseries	Calliandra calothyrsus (fodder)
	Leucaena trichandra (fodder)
	Tree Lucerne
	Grevillea robusta
	Croton spp.
	Sesbania sesban
Improved pasture and cattle management	Eucalyptus spp.
	Paddock
	Bush clearing
	Sowing with legumes
Manure management	Breeding
	Manure collection
	Compost
	Biogas digesters

Kenya: Implementation Strategy

- Through East Africa Dairy Development Project (EADD) and district extension services in 6 locations of Kaptumo division (27 000 people, 9000 ha)
 - As part of the Kaptumo Dairy Farmer Business Association (3 450 members)
 - Project extension officers trained → 5 Community Extension Service Providers →
 - **22 farmer trainers → 31 farmer groups**
 - 36 demo plots, 2 biogas digesters, 68 tree nurseries with > 300 000 seedlings
- **Total 4700 farmers trained (35% women)**

Farmer-to-farmer training approach:

Research showed that farmer trainers in the area can disseminate their knowledge up to 20 new farmers each month



Kenya: Implementation Strategy

EADD-MICCA project
Support

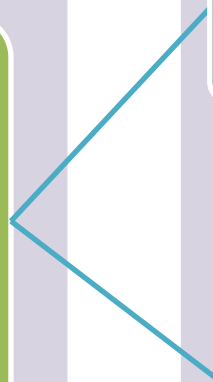


Trainings on
CSA practices



Trainings and workshops with the dairy association, the district officers of the Ministry of Agriculture and Livestock, and other development partners

5 Community
Extension service
providers and 22
Farmer trainers



31 Farmer
groups

Individual and
group tree
nurseries

36 Demo Plots

2 biogas digesters

4 700 farmers trained

exchange visits

68 tree nurseries

> 300 000 seedlings



MICCA Pilot Project in Tanzania

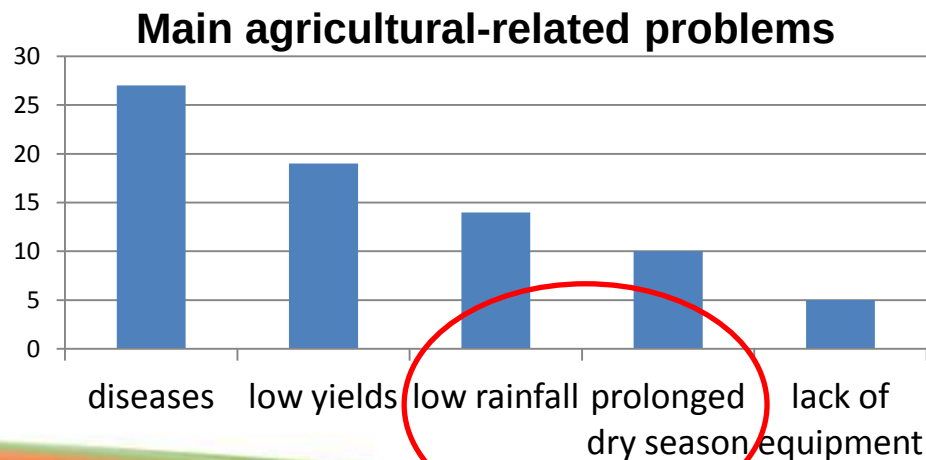
Cereal-based family farming in the highlands

*Combining conservation agriculture practices with agroforestry, improved cook-stoves, and soil and water conservation to **improve yield and livelihoods and reduce burning, erosion and deforestation.***



Tanzania: Baseline Results

- **Small land size:** median was 2 acres per farmer, but it ranged from 0.25 to 10 acres
- **Insecure Land Tenure:** 50% of farmers were renting lands (33% paying fees, many lands are clan-owned)
- 90% of farmers practice slash and burn agriculture
- **Climate change impacts** were perceived by 74% of households, e.g. prolonged dry season and crop failure -> resulting in food shortage



To cope with climate change, **57% of farmers didn't know what to do**, 16% planted cassava, and 6% practiced crop rotation.

Tanzania: Set of CSA practices



Practices	Sub-practices
Conservation agriculture	Minimum Tillage (after double digging)
	Mulching
	Intercropping with legume
	Crop Rotation
Agroforestry/ tree Planting/ tree nurseries	Multi-purpose trees: spice, fruit, woodfuel, fodder trees. E.g. cardamon, pepper, mango
Soil and Water Conservation and high value crops	Bench terraces (on 35 to 50% slope)
	Fanya Juu/ Chini (on 12-35% slope)
	Vegetative strips (on < 5% slope)
Improved Cooking Stoves	Household cooking stove
	Brew making stove

Tanzania: Implementation Strategy

- Through CARE HICAP project (Hillside Conservation Agriculture for Improved Livelihoods in the South Ulugurus) and district extension service in 3 wards and 15 villages (18 400 people, 16 800 ha)
- Training of trainers and **farmer-led extension approaches**
- **22 Farmer Field Schools** were trained
- **Demonstration plots** at Kolero Community Center and in farmer trainers' fields
- 736 **improved cooking stoves** built, inc. 50 for brew making
- A central and 11 satellite **tree nurseries (group and institutional)** with > 100 000 seedlings
- Exchange visits and agricultural exhibitions

Total 4 000 farmers trained on CSA

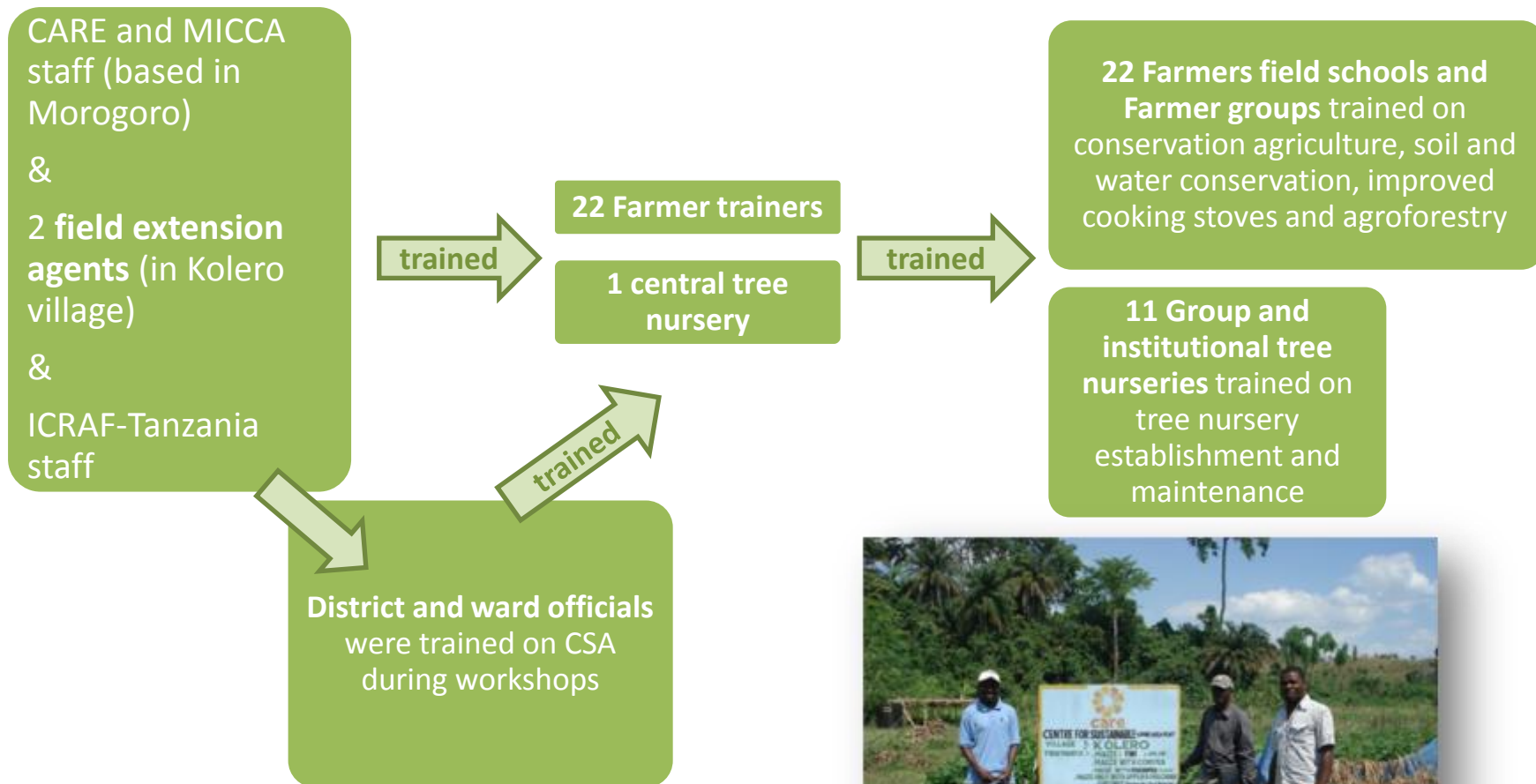
45% women



Brew making improved cook stove



Tanzania: Implementation Strategy



MICCA Pilot Projects- Lessons Learned

- Important to **target and tailor CSA** to site-specific farming system, socio-economic conditions and farmers needs.
- **Farmers** need to be engaged in the participatory planning of climate-smart agriculture and work jointly with technical specialists and extensionists to develop the set of locally-relevant CSA practices.
- **Sustainable extension** approaches are key, as adoption of CSA practices is highly influenced by trainings and farmer-to-farmer learning.

Smallholder farmers can be part of the solution to climate change, and increase food production and resilience while mitigating greenhouse gas emissions.





Thank you

For more information, please visit:

www.fao.org/climatechange/micca/pilots/

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